

AN ASYMPTOTIC DISTRIBUTION FOR THE DETERMINANT OF
A NON-CENTRAL B STATISTIC IN MULTIVARIATE ANALYSIS

D.J. de Waal

University of the Orange Free State

1. Introduction and summary

Let the random matrices $A_1(p \times p)$ and $A_2(p \times p)$ be independently distributed according to $W(\Sigma, m)$ and $W(\Sigma, n, \theta)$ respectively; i.e. A_1 is distributed according to a central Wishart with m degrees of freedom and A_2 is distributed according to a non-central Wishart with n degrees of freedom and non-centrality parameter θ of rank p . The distribution function of A_2 is defined as

$$1.1 \quad f(A_2) = w(A_2/\Sigma, n) \exp(\text{tr} - \frac{1}{2} \theta) {}_0F_1\left(\frac{n}{2}; \frac{1}{4} \theta \Sigma^{-1} A_2\right) |A_2|^{(n-p-1)/2}$$

$$w(A_2/\Sigma, n) = \frac{1}{\Gamma_p(\frac{n}{2}) |2\Sigma|^{(n/2)}} \exp(\text{tr} - \frac{1}{2} \Sigma^{-1} A_2) |A_2|^{(n-p-1)/2}$$

$$\text{where} \quad \Gamma_p(a) = \pi^{p(p-1)/4} \prod_{j=1}^p \Gamma(a + \frac{1-j}{2}).$$

The generalized hypergeometric function of matrix argument is defined as (Constantine (1963))

$$1.2 \quad {}_rF_q(a_1, \dots, a_r; b_1, \dots, b_q; S) = \sum_{k=0}^{\infty} \frac{(a_1)_K \dots (a_r)_K}{(b_1)_K \dots (b_q)_K} \frac{C_K(S)}{k!}$$

$$\text{where} \quad (a)_K = \prod_{j=1}^p (a + \frac{1-j}{2})_{K_j}$$

$$\text{and} \quad (a)_{K_j} = a(a+1)(a+2) \dots (a+K_j-1).$$

$K = (K_1, K_2, \dots, K_p)$, $K_1 \geq \dots \geq K_p \geq 0$, is the partition of k into not more than p parts such that

$$\sum_{j=1}^p K_j = k.$$

If a is such that the gamma function exists, $(a)_K$ can also be written as

$$(a)_K = \Gamma_p(a, K) / \Gamma_p(a)$$

$$\text{where} \quad \Gamma_p(a, K) = \pi^{\frac{1}{2}p(p-1)} \prod_{j=1}^p \Gamma(a + K_j + \frac{1-j}{2}).$$

$C_K(S)$ is a zonal polynomial (James (1961)) for which explicit formulas only exist in the cases $p=2$ and $S=I_p$, but has been tabulated (James (1964), James and Parkhurst) for values of $k=1(1)11$.

The purpose of this paper is firstly to derive the distribution of

$$L = (A_1 + A_2)^{-\frac{1}{2}} A_1 (A_1 + A_2)^{-\frac{1}{2}},$$

known as the generalized B statistic and secondly to derive an asymptotic distribution for $|L|$ and $|I-L|$. Due to the fact that

the zonal polynomials are not tabulated for higher values of k than 11, the accuracy of the approximation could not be found, but an example is given of the accuracy in the linear case, i.e. when the rank of θ is one.

2. The non-central Beta distribution

Theorem 1

If A_1 and A_2 are independently distributed according to $W(\Sigma, m)$ and $W(\Sigma, n, \theta)$ respectively, then the distribution function of L is defined as

$$\begin{aligned} 2.1 \quad f(L) = & \frac{1}{\Gamma_p(\frac{m}{2})} \frac{1}{\Gamma_p(\frac{n}{2})} \left(\frac{1}{|2\Sigma|} \right)^{\frac{m+n}{2}} \exp(\text{tr} - \frac{1}{2}\theta) |L|^{\frac{1}{2}(m-p-1)} \\ & \cdot |I-L|^{\frac{1}{2}(n-p-1)} \int_{T>0} \exp(\text{tr} - \frac{1}{2}\Sigma^{-1}T) |T|^{\frac{1}{2}(m+n-p-1)} \\ & \cdot {}_0F_1\left(\frac{n}{2}; \frac{1}{2}\theta \Sigma^{-1} T^{\frac{1}{2}} (I-L) T^{\frac{1}{2}}\right) dT \quad 0 < L < I \end{aligned}$$

Proof

Let $L = T^{-\frac{1}{2}} A_1 T^{-\frac{1}{2}}$, $T = A_1 + A_2$, in the joint distribution of A_1 and A_2 . The jacobian of the transformation is $J(A_1, A_2 \rightarrow L, T) = |T|^{\frac{1}{2}(p+1)}$. Integration w.r.t. T gives the distribution of L , but unfortunately this integral is unknown in the literature and a solution could not be found.

With reference to the work of Troskie (1966), this distribution is called the non-central multivariate Beta distribution of full rank.

Corollary 1

If L is distributed according to a non-central Beta distribution of full rank, then

$$2.2 \quad E|L|^h = \frac{\Gamma_p(\frac{m+n}{2}) \Gamma_p(\frac{m}{2}+h)}{\Gamma_p(\frac{m}{2}) \Gamma_p(\frac{m+n}{2}+h)} \exp(\text{tr} - \frac{1}{2}\theta) {}_1F_1\left(\frac{m+n}{2}; \frac{m+n}{2}+h; \frac{1}{2}\theta\right).$$

Proof

The following two integrals (Constantine (1963)) are necessary for the proof.

$$\begin{aligned} 2.3 \quad \int_0^I |L|^{a-\frac{1}{2}(p+1)} |I-L|^{b-\frac{1}{2}(p+1)} C_K(R(I-L)) dL \\ = \frac{\Gamma_p(a) \Gamma_p(b, K)}{\Gamma_p(a+b, K)} C_K(R), \quad R \text{ positive definite} \end{aligned}$$

$$\begin{aligned} 2.4 \quad \int_{T>0} \exp(\text{tr} - ST) |T|^{b-\frac{1}{2}(p+1)} C_K(RT) dT \\ = \Gamma_p(b, K) |S|^{-b} C_K(S^{-1}R), \quad S \text{ positive definite.} \end{aligned}$$

From 2.1

$$\begin{aligned} 2.5 \quad E|L|^h = & \frac{1}{\Gamma_p(\frac{m}{2})} \frac{1}{\Gamma_p(\frac{n}{2})} \frac{1}{|2\Sigma|^{\frac{m+n}{2}}} \exp(\text{tr} - \frac{1}{2}\theta) \\ & \cdot \int_0^I |L|^{\frac{1}{2}(m+2h-p-1)} |I-L|^{\frac{1}{2}(m+n-p-1)} \\ & \cdot \int_{T>0} \exp(\text{tr} - \frac{1}{2}\Sigma^{-1}T) |T|^{\frac{1}{2}(m+n-p-1)} \\ & \cdot {}_0F_1\left(\frac{n}{2}; \frac{1}{2}\theta \Sigma^{-1} T^{\frac{1}{2}} (I-L) T^{\frac{1}{2}}\right) dT dL. \end{aligned}$$

Integration w.r.t. L and T in this order using 2.3 and 2.4 respectively completes the proof.

Corollary 2

If L is distributed according to a non-central Beta distribution of full rank, then

$$2.6 \quad E|I-L| = \frac{\Gamma(\frac{m+n}{2}) \Gamma(\frac{n}{2}+h)}{\Gamma(\frac{n}{2}) \Gamma(\frac{m+n}{2}+h)} \exp(tr - \frac{1}{2}\theta) \cdot {}_2F_2\left(\frac{m+n}{2}, \frac{n}{2}+h; \frac{n}{2}, \frac{m+n}{2}+h; \frac{1}{2}\theta\right)$$

Proof

The proof is the same as that for Corollary 1.

3. Asymptotic distribution

The following approximation procedure will be used (Box (1949), Anderson (1958)).

Consider a function g of the form

$$3.1 \quad g(t) = C \left[\frac{\sum_{h=1}^q x_h^p}{\sum_{h=1}^q x_h^p} \right]^{-2it\rho} \prod_{j=1}^p \frac{\Gamma(\rho x_h(1-2it) + \beta_h + \epsilon_j)}{\Gamma(\rho x(1-2it) + \beta + \epsilon_j + \zeta_j)}$$

where C is a constant such that $g(0)=1$, ρ is a chosen constant ($0 < \rho < 1$), $\sum_{h=1}^q x_h = x$, $\beta_h = (1-\rho)x_h$, $\beta = (1-\rho)x$ and $i = \sqrt{-1}$.

Using the approximation for $\log \Gamma(x+h)$ the function can be written as

$$3.2 \quad g(t) = (1-2it)^{-\frac{1}{2}} f(1 + T_1(t) + T_2(t) + \dots + R)$$

where R is a remainder term that can be made as small as necessary.

$$3.3 \quad f = \sum_j ((q-1)(1-2\epsilon_j) + 2\zeta_j)$$

$$3.4 \quad T_1(t) = \omega_1((1-2it)^{-1} - 1)$$

$$T_2(t) = \omega_2((1-2it)^{-2} - 1) + \frac{1}{2}\omega_1^2((1-2it)^{-2} - 2(1-2it)^{-1} + 1),$$

ext.

$$3.5 \quad \omega_r = \frac{(-1)^{r+1}}{r(r+1)\rho^r} \sum_{j=1}^p \left(\sum_{h=1}^q \frac{B_{r+1}(\beta_h + \epsilon_j)}{x_h^r} - \frac{B_{r+1}(\beta + \epsilon_j + \zeta_j)}{x^r} \right)$$

Using the expressions for the Bernoulli polynomials, $B_r(y)$, it follows that

$$3.6 \quad \omega_1 = \frac{+1}{2\rho} \left[\sum_j \left\{ \left(\sum_h \frac{1}{x_h} - \frac{1}{x} \right) (\epsilon_j^2 - \epsilon_j + \frac{1}{6}) - \frac{1}{x} (\zeta_j^2 + 2\zeta_j \epsilon_j - \zeta_j) \right\} - (1-\rho)f \right]$$

3.1 is a very general form from which follows that when $q=1$

$$3.7 \quad g(t) = C \prod_{j=1}^p \frac{\Gamma(\rho x(1-2it) + \beta + \epsilon_j)}{\Gamma(\rho x(1-2it) + \beta + \epsilon_j + \zeta_j)}.$$

3.2 stays the same, but not

$$3.8 \quad f = 2 \sum_j \zeta_j \quad \text{and}$$

$$3.9 \quad \omega_1 = \frac{-1}{2\rho} \left(\frac{1}{x} \sum_j (\zeta_j^2 + 2\zeta_j \epsilon_j - \zeta_j) \right) + (1-\rho)f$$

Consider the hypothesis

$$H_0: \theta = 0$$

against the alternative

$$H_1: \theta \neq 0.$$

The following theorems can now be proved.

Theorem 2

An asymptotic distribution for the determinant of the generalized B statistic is given by

$$3.10 \quad P(-a \log |L| / H_0 \leq z) = P(\chi_{f_0}^2 \leq z) + O(m^{-2})$$

and

$$3.11 \quad P(-a \log |L| / H_1 \leq z) = \sum_{k=0}^{\infty} P(\chi_{f_k}^2 \leq z) P(k) \exp(tr - \frac{1}{2}\theta) \sum_{K=0}^{\infty} \sum_K [P(\chi_{f_k+2}^2 \leq z) - P(\chi_{f_k}^2 \leq z)] \omega_1 K \frac{C_K(\frac{1}{2}\theta)}{K!} + O(m^{-2})$$

where $a = \rho m$, $\rho = 1 + \frac{n-p-1}{2m}$, $f_k = np + 2k$,

$$\text{and } \omega_1 K = \frac{-1}{\rho} \left((1-\rho + \frac{n}{m})k + \frac{1}{m} (\sum_j K_j^2 - \sum_j K_j j) \right)$$

$$P(k) = \exp(tr - \frac{1}{2}\theta) \left(\frac{tr \frac{1}{2}\theta}{k!} \right)^k$$

Proof

Let $W = |L|^m/2$ and $M = -2 \log W$, then the characteristic function of ρM can be written in a form containing 3.7 by using 2.2, i.e.

$$3.12 \quad \begin{aligned} \phi_{\rho M}(t) &= E |L|^{-2it\rho} \left(\frac{m}{2} \right) \\ &= \exp(tr - \frac{1}{2}\theta) \sum_{k=0}^{\infty} \sum_K \left[\frac{\Gamma_p(\frac{m+n}{2}, K)}{\Gamma_p(\frac{m}{2})} \right. \\ &\quad \left. \cdot \frac{\Gamma_p(\frac{m}{2}(1-2it\rho))}{\Gamma_p(\frac{m}{2}(1-2it\rho) + \frac{n}{2}, K)} \right] \frac{C_K(\frac{1}{2}\theta)}{K!} \end{aligned}$$

The term in the square brackets is the same as 3.7 with $x = \frac{m}{2}$, $\epsilon_j = \frac{1-j}{2}$ and $\zeta_j = \frac{n}{2} + K_j$. Using the subscript k in 3.2 to denote the k -th term, 3.12 is approximately equal to

$$3.13 \quad \phi_{\rho M}(t) = \exp(tr - \frac{1}{2}\theta) \sum_K (1-2it)^{-\frac{1}{2}f_k} (1 + T_{1k}(t) + \dots + R_k) \cdot \frac{C_K(\frac{1}{2}\theta)}{K!}$$

where, from 3.8 and 3.9

$$f_k = np + 2k \text{ and}$$

$$3.14 \quad \omega_1 k = \frac{-1}{2\rho} \left[(1-\rho)f_k + \frac{n}{m} \left(\frac{p}{2}(n-p-1) + 2k \right) + \frac{2}{m} \left(\sum_j K_j^2 - \sum_j K_j j \right) \right].$$

But f_k is independent of K and, recalling that $\sum_K C_K(\frac{1}{2}\theta) = (\text{tr} \frac{1}{2}\theta)^k$,

3.13 can be written as

$$\begin{aligned} 3.15 \quad \phi_{\rho M}(t) = & \sum_{k=0}^{\infty} (1-2it)^{-\frac{1}{2}f_k} P(k) + \exp(\text{tr} \frac{1}{2}\theta) \cdot \\ & \sum_K \omega_{1K} ((1-2it)^{-\frac{1}{2}(f_k+2)} - (1-2it)^{-\frac{1}{2}f_k}) \cdot \\ & \frac{C_K(\frac{1}{2}\theta)}{k!} + O(m^{-2}) \end{aligned}$$

where $P(k) = \exp(\text{tr} \frac{1}{2}\theta) \frac{(\text{tr} \frac{1}{2}\theta)^k}{k!}$ (Poisson)

Because ω_r consists of order terms m^{-r} , all terms of the order m^{-2} and higher are ignored, but higher order terms may be taken into account if so desired.

Using the universe theorem (see Anderson (1958), page 206), it follows that

$$\begin{aligned} 3.16 \quad P(\rho M \leq z) = & \sum_{k=0}^{\infty} P(\chi_{f_k}^2 \leq z) P(k) + \exp(\text{tr} \frac{1}{2}\theta) \\ & \sum_{k=0}^{\infty} \sum_K \omega_{1K} (P(\chi_{f_k+2}^2 \leq z) - P(\chi_{f_k}^2 \leq z)) \frac{C_K(\frac{1}{2}\theta)}{k!} + O(m^{-2}) \end{aligned}$$

ρ can be found by letting $\omega_{10}=0$ in 3.14. Solving ρ in this equation it follows that

$$3.17 \quad \rho = 1 + \frac{1}{2m} (n-p-1).$$

3.16 now becomes

$$\begin{aligned} 3.18 \quad P(\rho M \leq z) = & P(-a \log |L| \leq z) \\ = & \sum_{k=0}^{\infty} P(\chi_{f_k}^2 \leq z) P(k) + \exp(\text{tr} \frac{1}{2}\theta) \sum_{k=1}^{\infty} \sum_K \omega_{1K} \\ & (P(\chi_{f_k+2}^2 \leq z) - P(\chi_{f_k}^2 \leq z)) \frac{C_K(\frac{1}{2}\theta)}{k!} + O(m^{-2}) \end{aligned}$$

where $a = \rho m = m + \frac{1}{2}(n-p-1)$ and

$$3.19 \quad \omega_{1K} = \frac{-1}{\rho} \left((1-\rho + \frac{n}{m}) \cdot k + \frac{1}{m} (\sum_j K_j^2 - \sum_j K_j j) \right).$$

Substituting $\theta=0$, i.e. the central case, 3.18 reduces to 3.10.

This completes the proof.

In Anderson (1958), page 208, the approximation of the central case is given with a remainder term $O(m^{-6})$.

The calculations may be simplified in the non-central case by letting any other ω_{1K} equal zero. One may choose, for example, ω_{1A} , where A is an integer.

A very important case is when θ is of rank one. Kshirsagar (1961) derived the distribution of L in this case, called the linear case. An asymptotic distribution for $|L|$ in the linear case follows easily from theorem 2 and is shown in the following corollary.

Corollary 3

An asymptotic distribution for the determinant of the generalized B statistic in the linear case is given by

$$3.20 \quad P(-\log |L| \leq z) = \sum_{k=0}^{\infty} P(\chi_{f_k}^2 \leq z) P(k) + \exp(tr - \frac{1}{2}\lambda) \sum_{k=0}^{\infty} \omega_{1k} \frac{(P(\chi_{f_k+2}^2 \leq z) - P(\chi_{f_k}^2 \leq z)) \frac{(\frac{1}{2}\lambda)^k}{k!}}{+0(m^{-2})}$$

where λ is the non-centrality parameter, $a = \rho m$, $f_k = np + 2k$,

$$3.21 \quad \rho = 1 + \frac{\frac{1}{2}np(n-p-1) + 2nA + 2A^2 - 2A}{m(np + 2A)},$$

and

$$3.22 \quad \omega_{1k} = \frac{-1}{\rho} \left[(1 - \rho + \frac{n}{m})(k - A) + \frac{1}{m}(k^2 - A^2) \right].$$

A is any chosen integer and

$$P(k) = \exp(tr - \frac{1}{2}\lambda) \frac{(\frac{1}{2}\lambda)^k}{k!}$$

Proof

When θ is of rank one, then $C_K(\frac{1}{2}\theta)$ becomes, say $(\frac{1}{2}\lambda)^k$, where λ is now the non-centrality parameter and ω_{1k} , given in 3.14, becomes

$$3.23 \quad \omega_{1k} = \frac{-1}{2\rho} \left[(1 - \rho)f_k + \frac{n}{m} \left(\frac{p}{2}(n-p-1) + 2k \right) + \frac{2}{m}(k^2 - k) \right].$$

Equations 3.21 and 3.22 now follow by letting $\omega_{1A} = 0$ and solving for ρ where A can be any integer. If $A = 0$, then ρ is the same as given in theorem 2.

Theorem 3

An asymptotic distribution for $|I-L|$ is given by

$$3.24 \quad P(-\log |I-L| \leq z) = P(\chi_f^2 \leq z) + 0(n^{-2})$$

where $a = n + \frac{1}{2}(m-p-1) + \frac{2A}{p}$

and $A = tr(\frac{1}{2}\theta)$, $f = mp$.

Proof

Let $V = |I-L| \frac{n}{2}$ and $M = -2 \log V$, then from 2.6 the characteristic function of ρM is

$$3.25 \quad \phi_{\rho M}(t) = \exp(tr - \frac{1}{2}\theta) \sum_{k=0}^{\infty} \frac{\Gamma_p(\frac{m+n}{2}, k) \Gamma_p(\frac{n}{2}(1-2it\rho), k)}{\Gamma_p(\frac{n}{2}, k) \Gamma_p(\frac{n}{2}(1-2it\rho) + \frac{m}{2}, k)} \cdot \frac{C_K(\frac{1}{2}\theta)}{k!}$$

The term in square brackets is now of the form 3.7 with

$$x = \frac{n}{2}, \quad \epsilon_j = K_j + \frac{1-j}{2} \quad \text{and} \quad \zeta_j = \frac{m}{2}.$$

Therefore

$$3.26 \quad f = mp \quad \text{and}$$

$$3.27 \quad \omega_{1k} = \frac{-1}{2\rho} \left(\frac{m}{n} \left(\frac{mp}{2} - \frac{p^2}{2} + 2k - \frac{p}{2} \right) + (1-\rho)f \right)$$

Choosing $\omega_{1A} = 0$ (A any integer) and solving for ρ , it follows that

$$3.28 \quad \rho = 1 + \frac{2A}{np} + \frac{m-p-1}{2n}.$$

Therefore

$$3.29 \quad \omega_{1k} = \frac{m}{a} (A - k), \quad a = np.$$

It is important to note that f is independent of k and ω_{1k} independent of K . Therefore substituting 3.2 for the term in square brackets in 3.25, ignoring terms of order n^{-2} and higher, it follows that

$$3.30 \quad \phi_{pM}(t) = \exp(tr - \frac{1}{2}\theta) \sum_{k=0}^{\infty} \frac{\Sigma(1-2it)^{-\frac{1}{2}f}(1+T_{1k}(t))}{k!} + O(n^{-2})$$

$$= (1-2it)^{-\frac{1}{2}f} + ((1-2it)^{-\frac{1}{2}(f+2)} - (1-2it)^{-\frac{1}{2}f}).$$

$$\frac{m}{a}(A - tr\frac{1}{2}\theta) + O(n^{-2}).$$

Taking $A = tr(\frac{1}{2}\theta)$, the second term of 3.30 is equal to zero. By using the inverse theorem the proof is completed.

It is interesting to note that when $\theta=0$, the central case, 3.24 remains the same except for ρ which becomes

$$\rho = 1 + \frac{m-p-1}{2n}.$$

4. An example of the power of the test given in corollary 3

In the case $p=2$ exact values are available of the power function of $|L|$ (linear case) for various values of λ , m and n (Roy (1965)). The asymptotic values of the test given in 3.20 can thus be compared with the exact values. An extract from a number of tables of the power function at the 5% level which have been computed for different parameter values are given in tables 1 and 2. Table 1 gives a comparison of the exact values of the power function with the approximated values for $p=2$, $n=10$, $m=50, 100, \infty$ and $\lambda=2, 6, 10, 16, 28, 40$. Table 2 gives a comparison of the exact values, the chi-square approximation, the gamma approximation and the Jacobi-series approximation (Roy (1965)) for $p=2$, $n=4$, $m=50, 100$ and $\lambda=2, 10$.

From table 1 it is seen that the approximation is fairly accurate. In table 2 the approximation seems to be better than the gamma approximation and the Jacobi-series approximation for $\lambda=10$.

TABLE 1

A Comparison of exact values and chi-square approximation of the power function of $|L|$ for $p=2$ and $n=10$ at 5% level

λ	$m=50$		$m=100$		$m=\infty$	
	Exact	χ^2	Exact	χ^2	Exact	χ^2
2	0.0874	0.0841	0.0913	0.0904	0.0961	0.0956
6	.1910	.1808	.2092	.2063	.2319	.2293
10	.3208	.3053	.3574	.3532	.4019	.3970
16	.5257	.5158	.5835	.5813	.6483	.6415
28	.8283	.8472	.8797	.8853	.9240	.9200
40	.9545	.9712	.9765	.9805	.9828	.9889

TABLE 2

A Comparison of exact values, chi-square approximation, gamma approximation and Jacobi-series approximation of the power function of $|L|$ for $p=2$ and $n=4$ at 5% level

m=50				
λ	Exact	χ^2	Gamma	Jacobi
2	0.1209	0.1196	0.1212	0.1201
10	.5225	.5201	.5033	.5017
m=100				
2	.1257	.1254	.1261	.1244
10	.5545	.5539	.5469	.5388

Acknowledgment

The author wishes to express his thanks to Prof. C.G. Troskie for valuable suggestions, Prof. A. Reitsma, who made the computations possible and Mr L.J.C. Bootha for the programming of the material.

REFERENCES

- ANDERSON, T.W. (1958). *An Introduction to Multivariate Statistical Analysis*. Wiley, N.Y.
- BOX, G.E.P. (1949). A general distribution theory for a class of likelihood criteria. *Biometrika*, 36, 317-346.
- CONSTANTINE, A.G. (1963). Some non-central distribution problems in multivariate analysis. *Ann. Math. Statist.*, 34, 1270-1285.
- JAMES, A.T. (1961). Zonal polynomials of the real positive definite symmetric matrices. *Ann. Math.*, 74, 456-469.
- JAMES, A.T. (1964). Distribution of matrix variates and latent roots derived from normal samples. *Ann. Math. Statist.*, 35, 475-501.
- JAMES, A.T. and PARKHURST, A. Real zonal polynomials of orders 7 through 11. To be submitted to *Ann. Math. Statist.*
- KSHIRSAGAR, A.M. (1961). The non-central multivariate Beta distribution. *Ann. Math. Statist.*, 32, 104-111.
- ROY, J. (1965). Power of the likelihood-ratio test used in analysis of dispersion. *Multivariate Analysis*. (Krishnaiah (1966)). Academic Press, N.Y.
- TROSKIE, C.G. (1966). Nie-sentrale meerveranderlike Beta-verdelings. *Tydskrif vir Natuurwetenskappe*, 6, 58-71.